Step 1: Create a role name - `AWSCustomRoleForEC2` with `CloudWatchFullAccess` permission. Assign the above role to your running AWS EC2 instance.

Step 2: Download and install the Cloudwatch agent in your EC2 instance

```
wget
https://s3.amazonaws.com/amazoncloudwatch-agent/ubuntu/amd64/latest/amazon-clo
udwatch-agent.deb

sudo dpkg -i -E ./amazon-cloudwatch-agent.deb
```

Step 3: Launch the Cloudwatch wizard

```
sudo
/opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-config-wizard
```

Choose these settings

```
=============================================================

= Welcome to the AWS CloudWatch Agent Configuration Manager =

=============================================================

On which OS are you planning to use the agent?

1. linux

2. windows

default choice: [1]:


Trying to fetch the default region based on ec2 metadata...

Are you using EC2 or On-Premises hosts?

1. EC2

2. On-Premises

default choice: [1]:
```

```
Which user are you planning to run the agent?

1. root

2. cwagent

3. others

default choice: [1]:


Do you want to turn on StatsD daemon?

1. yes

2. no

default choice: [1]:


Which port do you want StatsD daemon to listen to?

default choice: [8125]


What is the collect interval for StatsD daemon?

1. 10s

2. 30s

3. 60s

default choice: [1]:

3


What is the aggregation interval for metrics collected by StatsD daemon?

1. Do not aggregate

2. 10s

3. 30s

4. 60s

default choice: [4]:
```

```
4

Do you want to monitor metrics from CollectD?
1. yes
2. no
default choice: [1]:
2

Do you want to monitor any host metrics? e.g. CPU, memory, etc.
1. yes
2. no
default choice: [1]:

Do you want to monitor cpu metrics per core? Additional CloudWatch charges may apply.
1. yes
2. no
default choice: [1]:
2

Do you want to add ec2 dimensions (ImageId, InstanceId, InstanceType, AutoScalingGroupName) into all of your metrics if the info is available?
1. yes
2. no
default choice: [1]:
2
```

```
Would you like to collect your metrics at high resolution (sub-minute
resolution)? This enables sub-minute resolution for all metrics, but you can
customize for specific metrics in the output json file.

1. 1s

2. 10s

3. 30s

4. 60s

default choice: [4]:

4


Which default metrics config do you want?

1. Basic

2. Standard

3. Advanced

4. None

default choice: [1]:


Current config as follows:
```



```
{

    "agent": {

        "metrics_collection_interval": 60,

        "run_as_user": "root"

    },

    "metrics": {

        "metrics_collected": {

            "disk": {

                "measurement": [
```

```
                    "used_percent"
                ],
                "metrics_collection_interval": 60,
                "resources": [
                    "*"
                ]
            },
            "mem": {
                "measurement": [
                    "mem_used_percent"
                ],
                "metrics_collection_interval": 60
            },
            "statsd": {
                "metrics_aggregation_interval": 60,
                "metrics_collection_interval": 60,
                "service_address": ":8125"
            }
        }
    }
}
```

```
Are you satisfied with the above config? Note: it can be manually customized
after the wizard completes to add additional items.
1. yes
2. no
default choice: [1]:
```

```
Do you have any existing CloudWatch Log Agent
(http://docs.aws.amazon.com/AmazonCloudWatch/latest/logs/AgentReference.html)
configuration file to import for migration?

1. yes

2. no

default choice: [2]:


Do you want to monitor any log files?

1. yes

2. no

default choice: [1]:

2


Saved config file to /opt/aws/amazon-cloudwatch-agent/bin/config.json
successfully.
```

Step 4: Start the Cloudwatch agent

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a
fetch-config -m ec2 -s -c
file:/opt/aws/amazon-cloudwatch-agent/bin/config.json
```

Tip: Command to check status of the Cloudwatch agent

```
service amazon-cloudwatch-agent status
```

`To update config.json file as autoscaling base with memory utilization , update the below config.json file.`

```
Vi /opt/aws/amazon-cloudwatch-agent/bin/config.json
```

```
# Update the below json file.

{

    "agent": {

        "metrics_collection_interval": 60,

        "run_as_user": "root"

    },

    "metrics": {

        "namespace": "CWAgent",

        "aggregation_dimensions": [["AutoScalingGroupName"]],

        "append_dimensions": {

            "AutoScalingGroupName": "${aws:AutoScalingGroupName}",

            "InstanceType": "${aws:InstanceType}"

        },

        "metrics_collected": {

            "disk": {

                "measurement": [

                    "used_percent"

                ],

                "metrics_collection_interval": 60,

                "resources": [

                    "*"

                ]

            },

            "mem": {

                "measurement": [

                    "mem_used_percent"

                ],
```

```
                "metrics_collection_interval": 60
            }
        }
    }
}
```

Run Below command to restart cloudwatch agent.

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a fetch-config -m ec2 -s -c file:/opt/aws/amazon-cloudwatch-agent/bin/config.json
```

Run Below command to check cloudwatch agent status.

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -m ec2 -a status
```